HW 3

Thursday, February 18, 2021 4:01 PM

2.1. Evaluate the following expressions.

(a) $(\lambda x, x^2 + x + 1)(3)$
(b) $(\lambda x, x^2 + x + 1)(s)$
(c) $(\lambda x, x^2 + x + 1)(s^2 + s + 1)$
(d) $(\lambda x, 5)(t)$
(e) $(\lambda x, 5)(t + 1)$
(f) $(\lambda x, \cos(x) + \sin(y))(1)$
(g) $(\lambda y, \cos(x) + \sin(y))(1)$
(h) $\lambda x, (\lambda y, \cos(x) + \sin(y))(1)$
(i) $(\lambda x, (\lambda y, \cos(x) + \sin(y))(1))(2)$
(j) $(\lambda T, T \text{ is isosceles})(ABC)$
(k) $(\lambda T, T \text{ is isosceles})(DEF)$

a) $(\lambda x, x^2+x+1)(3)$
$3x^2+3+1$

b) $(\lambda x, x^2+x+1)(s)$
s^2+s+1

c) $(\lambda x, x^2+x+1)(s^2+s+1)$
$(s^2+s+1)^2+(s^2+s+1)+1$

d) $(\lambda x, 5)(t)$
5

e) $(\lambda x, 5)(t+1)$
5

f) $(\lambda x, \cos(x)+\sin(y))(1)$
$\cos(1)+\sin(y)$

g) $(\lambda y, \cos(x)+\sin(y))(1)$
$\cos(x)+\sin(1)$

h) $(\lambda x)(\lambda y, \cos(x)+\sin(y))(1)$
$\lambda x, \cos(x)+\sin(1)$

i) $(\lambda x, (\lambda y, \cos(x)+\sin(y))(1))(2)$
$\lambda x, \cos(x)+\sin(1)(2)$
$\cos(2)+\sin(1)$

j) $(\lambda T, T \text{ is isosceles})(ABC)$
ABC is isosceles

k) $(\lambda T, T \text{ is isosceles})(DEF)$
DEF is isosceles

2.2. Let f be $\lambda n, 2 \cdot n$ and let g be $\lambda n, n + 2$. Evaluate the following.

(a) $f(k)$
(b) $g(k)$
(c) $f(g(k))$
(d) $g(f(k))$
(e) $f(f(k))$
(f) $g(g(k))$
(g) $f(g(k + 1))$
(h) $g(f(k + 1))$
(i) $f(f(k + 1))$
(j) $g(g(k + 1))$

$f(n) = \lambda n, 2 \cdot n$ $\quad g(n) = \lambda n, n+2$

(a) $(\lambda n, 2\cdot n)(k)$
$2 \cdot k$

(b) $(\lambda n, n+2)(k)$
$k+2$

c) $(\lambda n, 2\cdot n)((\lambda n, n+2)(k))$
$2\cdot(k+2)$

d) $g(f(k)) = (\lambda n, n+2)((\lambda n, 2\cdot n)(k))$
$(2\cdot k)+2$

e) $f(f(k)) = (\lambda n, 2\cdot n((\lambda n, 2\cdot n)(k))$
$2\cdot 2(k)$

f) $g(g(k) = (\lambda n, n+2)((\lambda n, n+2)(k))$
$(k+2)+2$

g) $f(g(k+1)) = (\lambda n, 2\cdot n)((\lambda n, n+2)(k+1))$
$2\cdot((k+1)+2)$

h) $g(f(k+1)) = (\lambda n, n+2)((\lambda n, 2\cdot n)(k+1))$
$(2\cdot(k+1))+2$

i) $f(f(k+1)) = (\lambda n, 2\cdot n)((\lambda n, 2\cdot n)(k+1))$
$2\cdot(2\cdot(k+1))$

j) $g(g(k+1) = (\lambda n, n+2)((\lambda n, n+2)(k+1)$
$((k+1)+2)+2$

2.3. Suppose we have the following named lambda expressions.

$$\lambda W, \lambda V, WV, VW \vdash W \qquad (R_1)$$
$$\lambda W, \lambda V, W, V \vdash W \bullet V \qquad (R_2)$$
$$\lambda W, \lambda V, WV \bullet \vdash W \bigcirc \qquad (R_3)$$

Evaluate the following.

(a) $(R_1(\bullet))(\bigcirc)$
(b) $(R_2(\bullet))(\bigcirc)$
(c) $(R_3(\bullet))(\bigcirc)$
(d) $R_1(\bullet\bigcirc\bullet)$
(e) $R_2(\bullet\bigcirc\bullet)$
(f) $R_3(\bullet\bigcirc\bullet)$

$$\lambda W, \lambda V, (WV, VW \vdash W)$$

So that, for example, substituting $\bigcirc$ for W and $\bullet$ for V produces the rule

$$(\lambda W, \lambda V, (WV, VW \vdash W))(\bigcirc)(\bullet) = (\lambda V, (\bigcirc V, V\bigcirc \vdash \bigcirc))(\bullet)$$
$$= (\bigcirc\bullet, \bullet\bigcirc \vdash \bigcirc)$$

$\lambda W, \lambda V(WV, VW \vdash W)$

a) $(R_1(\bullet))(\bigcirc)$
$(\lambda W, \lambda V(WV, VW \vdash W))(\bullet)(\bigcirc)$
$\lambda \bigcirc, \lambda V(\bigcirc V, V\bigcirc \vdash \bigcirc)(\bullet)$
$\lambda \bigcirc, \lambda \bullet(\bigcirc\bullet, \bullet\bigcirc \vdash \bigcirc)$
$\bigcirc$

b) $(R_2(\bullet))(\bigcirc) = (\lambda W, (\lambda V, W, V \vdash W\bullet V(\bullet))(\bigcirc))$
$(\lambda \bigcirc, \lambda V, (\bigcirc, V \vdash \bigcirc\bullet V)(\bullet)$
$\lambda \bigcirc, \lambda \bullet, (\bigcirc, \bullet \vdash \bigcirc\bullet\bullet)$
$\boxed{\bigcirc\bullet\bullet}$

d) $R_1(\bullet\bigcirc\bullet) = (\lambda W, \lambda V, WV, VW \vdash W)(\bullet\bigcirc\bullet)$
$\lambda \bullet\bigcirc\bullet, \underline{\lambda V}, \bullet\bigcirc\bullet\underline{V}, \underline{V}\bullet\bigcirc\bullet \vdash \bullet\bigcirc\bullet$

e) $R_2(\bullet\bigcirc\bullet) = (\lambda W, \lambda V, W, V \vdash W\bullet V)(\bullet\bigcirc\bullet)$
$\lambda \bullet\bigcirc\bullet, \underline{\lambda V}, \bullet\bigcirc\bullet, \underline{V}, \vdash \bullet\bigcirc\bullet\bullet\underline{V}$

f) $R_2(\bullet\bigcirc\bullet) = (\lambda W, \lambda V, WV\bullet \vdash W\bigcirc)(\bullet\bigcirc\bullet)$

$\boxed{0\bullet\bullet}$

(c) $(R_3(\bullet))(0) = \lambda w.(\lambda v.(wv\bullet + w0)(\bullet))(0)$

$\lambda 0.((\lambda v.(0v\bullet + 00)(\bullet))$

$\lambda 0.\lambda \bullet.0\bullet\bullet + 00$

$\boxed{00}$

f) $R_3(\bullet 0 \bullet) = ((\lambda w.\lambda v.wv\bullet + w0)(\bullet 0 \bullet))$

$\lambda \bullet 0 \bullet.\underline{\lambda v}.\bullet 0 \bullet \underline{v} \bullet + \bullet 0 \bullet 0$